MagicPaint: Operate Anything for Image Inpainting with Diffusion Model

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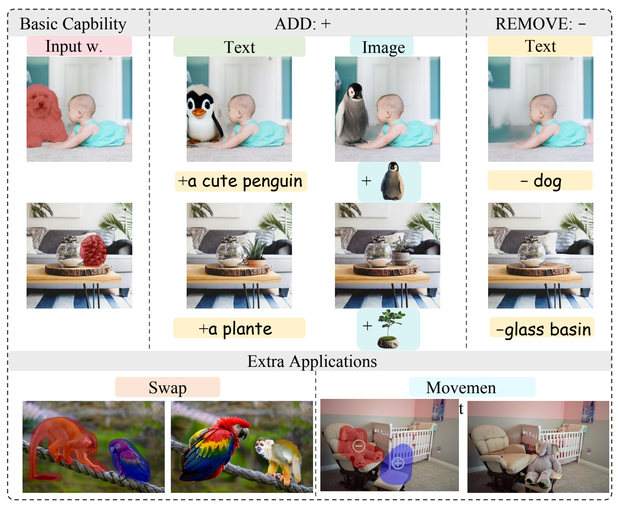
Qinhong Yang, Dongdong Chen, Qi Chu, Tao Gong*, Qiankun Liu, Zhentao Tan, Xulin Li, Huamin Feng, Nenghai Yu

Motivation: Why MagicPaint?

Background

Leveraging pre-trained generative priors like Stable Diffusion, modern image inpainting has evolved from simple background repair to sophisticated content synthesis, utilizing diffusion-based architectures and multimodal inputs (text and images) to achieve seamless region restoration, high-fidelity object removal, and condition-guided synthesis within a unified pipeline.



Challenge

Task Incompatibility

Current models are "one-trick ponies"—specializing in either addition, removal, or inpainting, but failing to handle them unified in a single framework.

Conflicting Objectives

When removing objects, generative priors often "hallucinate" new content or leave artifacts because the model cannot distinguish between adding and restoring objectives.

Coarse Guidance

Existing methods rely on high-level text prompts (semantic only), which lack precise spatial control and fine-grained operational intent.

Data Scarcity

A significant lack of large-scale, high-quality, multi-modal (text + image) datasets that support diverse editing tasks simultaneously.

Our work

We investigate a single model called MagicPaint handling Context-driven inpainting, text-guided addition, reference image-guided addition, and text-guided removal harmoniously.

Key Contributions

MagicPaint Model:

MagicMask: Pixel-level spatial guidance for task differentiation and

MMToken: Decouples operation intent from modality.

MMData Dataset: 2.1M high-quality editing pairs.

Link

Multi-Task Method: MagicMask & MMToken

MagicPaint framework

MagicPaint addresses conflicting optimization objectives in unified inpainting through fine-grained guidance at both the semantic and spatial levels.

MMToken: Dual-Dimension Decoupling

The MMToken module decouples operation intent from target content by utilizing three learnable tokens: <Add>, <Remove>, and <Inpaint> and multimodal inputs (text and image), aligning them into a unified representation space to effectively distinguish between diverse inpainting tasks.

MagicMask: Operation-Aware Spatial Guidance

MagicMask transforms the standard mask into an operation-specific guide by encoding task intent through signed values and modulating latent inputs to provide either a full image for precise object removal or a masked image to compel content synthesis during addition and inpainting tasks.

$$x_t' = [z_t, m_o, x_0 \odot (1 - m_o^E)],$$

MaskGen & Unified Optimization

To achieve pixel-level precision, we introduce **MaskGen**, a dedicated sub-module integrated with a joint supervision strategy.

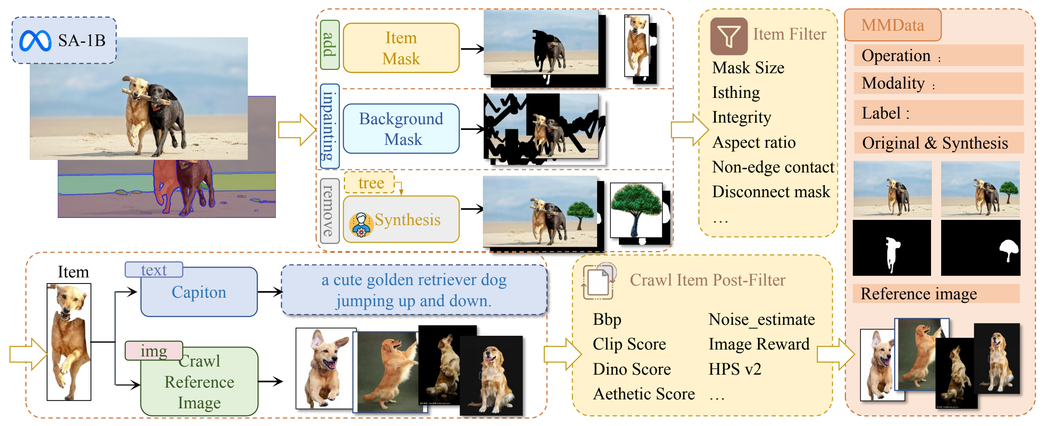
$$m' = \text{MaskGen}(\hat{x_0}, t, \tau(c')), \qquad \mathcal{L}_m = (\mathcal{L}_{bce}(m_{gt}, m') + \mathcal{L}_{dice}(m_{gt}, m'))/2$$

$$\mathcal{L} = \mathcal{L}_\epsilon + \lambda \cdot \mathcal{L}_m$$

New Dataset: MMDATA

Dataset

We introduce MMData, a large-scale, high-quality inpainting dataset containing 2.1M high-quality editing samples. It captures diverse inpainting scenarios and supports both text and image input modalities.



Advantages

Unprecedented Scale & Resolution

✓ 2.1M+ high-quality image editing pairs.

✓ Support for high-resolution image processing, filling the gap in large-scale region-specific inpainting.

Multi-Modal & Multi-Task Versatility

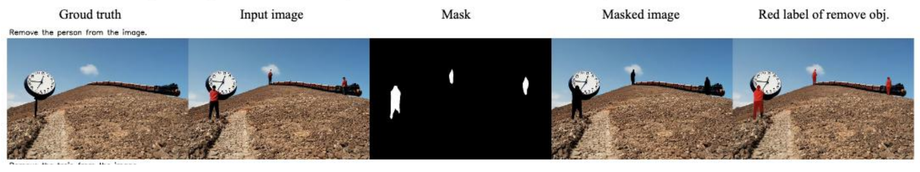
✓ 4-in-1 Tasks: Context-driven inpainting, text-guided addition, reference image-guided addition, and text-guided removal.

✓ Multi-modality: Supports both textual prompts and visual reference images.

Realistic Object Removal

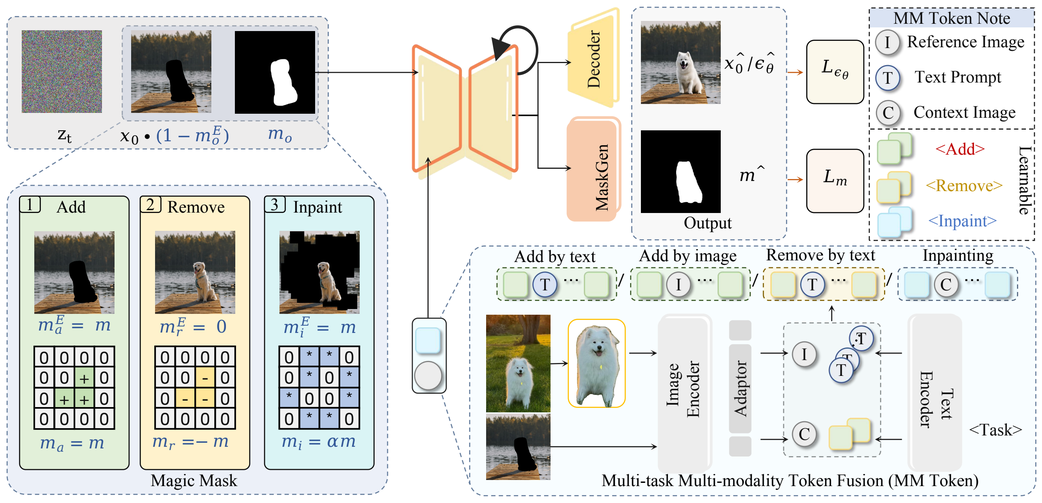
✓ Captures complex physical interactions including realistic shadows and occlusions.

✓ Unique support for Multi-turn removal within the same scene, maintaining high fidelity and consistency.



Framework and Experiments

Framework



Experiments

Module	Task	Add by Text		Add by Image		Remove by text		Inpaint	
		L-FID↓	CLIP↑	FID ↓	LPIPS ↓	REMOVE↑	FID↓	FID ↓	LPIPS ↓
SD-I	ori	5.65	24.81	-	-	0.768	86.0	3.03	0.1312
SD-I	remove text	9.13	18.77	-	-	0.787	71.7	3.00	0.1307
SD-I (+VE)	all	7.29	22.23	24.98	0.2761	0.743	112.3	2.71	0.1339
+ MM Token		7.07	25.86	19.43	0.1898	0.783	74.9	2.67	0.1134
+ MagicMask		6.84	25.25	19.98	0.2375	0.796	72.1	2.46	0.1227
MagicPaint†		**5.53**	**27.73**	**14.98**	**0.1691**	**0.817**	**65.8**	**2.19**	**0.0967**

Conclusion

- We propose MagicPaint, a unified diffusion-based model designed to effectively handle diverse inpainting tasks, including object addition, removal, swapping, and movement, within a single framework.
- We contribute a large-scale dataset named MMData, consisting of 2.1 million samples, to support the study of multimodal inpainting. Compared to existing datasets, MMData stands out for its high-quality training signals across diverse scenarios, particularly significantly improving performance in object removal.
- In MagicPaint, we semantically disentangle operation types from target content and encode operation intent directly into the mask to provide operation-aware spatial supervision, alleviating the conflicts inherent in multi-task inpainting.